

CHE 205 MidTerm Exam 2

April 8th, 2020

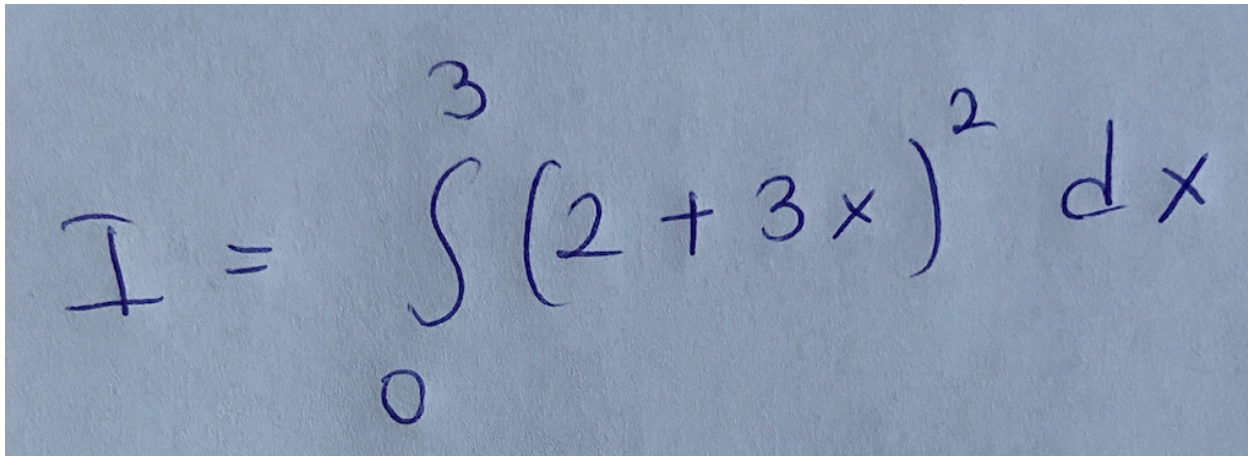
NAME: _____

- Exam is open-book and open-note. You are allowed to use internet, however, you are NOT allowed to talk to, text or email anybody during the exam.
- You need to submit your excel spreadsheet on BB before 11.30am. Make sure you save your file frequently and save as xlsx to be able to save your VBA codes. If you do not save as xlsx, I will not be able to see your VBA code and you will lose points on those problems.
- Make sure you keep a copy of your work on your computer.
- Highlight your answers and make sure to label constants, initial values and equations. I need to follow your logic.
- Save each problem as different sheet in the same excel file. Screenshot your VBA code and paste it into excel.
- Good luck!

Question 1 (40 points):

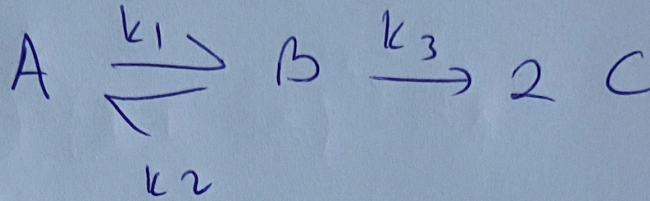
Evaluate the following integral using

- a. Trapezoidal rule. Use different increments.
- b. Gauss quadrature. Use 2-, 3- and 4-point formulas and compare results.



A photograph of a handwritten equation in blue ink on a light blue background. The equation is
$$I = \int_0^3 (2 + 3x)^2 dx$$

Question 2 (60 points):



$$\frac{dC_A}{dt} = -k_1 C_A + k_2 C_B$$

$$\frac{dC_B}{dt} = k_1 C_A - k_2 C_B - k_3 C_B$$

$$\frac{dC_C}{dt} = 2 k_3 C_B$$

$$k_1 = 1 \text{ min}^{-1} \quad k_2 = 0.5 \text{ min}^{-1} \quad k_3 = 2 \text{ min}^{-1}$$

$$C_A = 10 \text{ mol/L} \quad C_B = C_C = 0 \text{ mol/L}$$

- Solve this ODE-IVP using Euler method and Excel
- Solve this ODE-IVP using Euler method and VBA
- Solve this ODE-IVP using second order RK method and Excel
- Solve this ODE-IVP using second order RK method and VBA
- Solve this ODE-IVP using fourth order RK method and Excel
- Solve this ODE-IVP using fourth order RK method and VBA